

RAG Grounding Evaluation Report

Week 04 — RAG: Automation Over Your Own Docs

1. Method

An evaluation set of 18 question-answer pairs was built based on the content of the VaultX AI Internship Week 01-04 documents (the ingested corpus). Of these, 16 questions have answers present in the documents, and 2 are designed to test correct refusal (should_refuse = true), where the expected behavior is that the RAG system responds 'Answer not found in documents.' rather than hallucinating an answer.

Each question was run through the RAG pipeline (retrieve_chunks + generate_answer), and both the generated answer and the retrieved source chunks were logged. Each answer was manually reviewed and marked on two dimensions:

- **Correct** — whether the answer matches the expected answer from the source documents.
- **Grounded** — whether the answer's content is actually supported by the retrieved chunks (as opposed to being invented by the model).

The evaluation was run twice: once with the baseline retrieval setting (k=3), and once with an increased retrieval setting (k=5), to test whether retrieving more chunks improves recall on questions that initially failed.

2. Results Summary

Metric	k=3 (Baseline)	k=5 (After Improvement)
Total questions	18	18
Correct answers	13 / 18 (72.2%)	15 / 18 (83.3%)
Grounded (when an answer was given)	13 / 13 (100%)	15 / 15 (100%)
Correct refusals (should_refuse cases)	2 / 2 (100%)	2 / 2 (100%)
Hallucinations (ungrounded answers)	0	0

Increasing k from 3 to 5 improved accuracy from 72.2% to 83.3%, resolving 2 of the 5 original failures. Grounding remained at 100% in both settings — whenever the system produced an answer (as opposed to refusing), that answer was fully supported by the retrieved chunks. No hallucinations were observed at either setting.

3. Failure Analysis

Three questions remained unanswered even after increasing k to 5:

#	Question	Expected Answer	Likely Cause
3	How many terms were required in the Week 1 glossary?	9 terms	The fact was not fully captured within any single retrieved chunk, likely due to how the deliverables table was split during chunking.

12	What needed to be added to handle failures in Week 3?	Error branches, retries, and fallback	Same pattern — the specific list item may be split across a chunk boundary, keeping it out of all retrieved chunks.
15	How many QA pairs were required for Week 4 evaluation?	15-20 QA pairs	The relevant sentence sits inside a task list that may be embedded with lower semantic similarity to the question wording.

In all three remaining failures, the system correctly avoided hallucinating an answer and instead responded 'Answer not found in documents.' This is the safe failure mode — the system did not invent facts, it simply could not retrieve the relevant chunk. The likely root cause is chunking rather than the retrieval count (k), since increasing k from 3 to 5 did not resolve these three cases.

4. Recommendations for Further Improvement

- **Smaller chunk size or higher overlap** — the remaining failures suggest that specific facts near deliverables/task-list boundaries may be getting split across chunks. Reducing chunk size or increasing overlap could keep these facts intact within a single chunk.
- **Reranking** — a reranking step after initial retrieval could help surface the correct chunk even when it is not in the top-k by raw similarity score.
- **Query rewriting** — reformulating the question before retrieval (e.g., adding keywords like 'deliverable', 'requirement') could improve match quality for structured list-based content.

5. Conclusion

The RAG system demonstrates strong grounding behavior: it never hallucinated an answer across 18 test questions, and correctly identified when information was genuinely absent from the documents (both should-refuse test cases passed at 100%). The main area for improvement is recall — a small number of facts embedded in structured list content are not being retrieved. Increasing k from 3 to 5 improved accuracy from 72.2% to 83.3%, and further gains are expected from chunking and reranking improvements.